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Bone Tumor Detection using Faster R-CNN

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***Abstract*— Bone tumors lead to bone cancer which can be prevented by precise detection. A Computer-Aided Diagnosis system helps to diagnose critical and subtle situations precisely as many hospitals lack experienced doctors. A tumor detection model using the Faster Region-based Convolutional Neural Network (R-CNN) is proposed which can detect tumors from the X-ray images. This work comprises two tasks: classification or recognition and localization. A Convolutional Neural Network (CNN) model is designed to classify the tumor which acts as the backbone of the Faster R-CNN model. During the training of the CNN model, several techniques like augmentation, early stopping, etc. are used to avoid underfitting and overfitting issues. A required number of layers of the trained CNN model are used in the Region Proposal Network (RPN) through the Transfer learning method. The RPN locates the position of tumors by bounding boxes. The dataset used to train the model consists of a total of 3522 images, of which 2283 are tumor-containing images and the remaining amount of the dataset are non-tumor images. 97.43% validation accuracy and 98.73 % test accuracy in the CNN model for classification and 96.28% Mean Average Precision (mAP) in Faster R-CNN have been achieved.**

Keywords*— *bone tumor detection, CNN, X-ray, transfer learning, RPN.

I. Introduction

The number of bone tumors and cancer-affected patients is increasing day by day. The World Health Organization classifies bone tumors as either primary or secondary types. Tumors, which may arise in various body areas, can be benign or malignant and exhibit diverse appearances. Primary bone tumors often present subtle symptoms, such as bone fractures, swelling, new bone growth, or swelling in adjacent soft tissues [1]. As the initial stage of bone cancer, a bone tumor requires early detection to enable effective treatment and potentially prevent cancer progression. Accurate detection is essential, with imaging techniques like radiographs, computed tomography, magnetic resonance imaging, and nuclear scintigraphy playing a key role in identifying bone abnormalities. X-ray image analysis is one of the most common and affordable screening tools for diagnosing bone abnormalities. Cancer-affected bones display unique X-ray absorption rates compared to healthy bones, producing images with ragged surfaces, tumors, or moth-eaten patterns that reflect the permeative, geographic, and irregular bone destruction characteristic of cancer [2][3]. Texture variations in affected areas may indicate the stage and severity of bone cancer, assisting doctors in assessing disease progression and treatment prognosis [3]. Digital X-ray processing, leveraging advanced technology, can aid in more accurately detecting tumors and abnormalities, particularly in sensitive regions that are difficult for specialists to diagnose visually [4]. Various methods exist for processing digital X-rays, with many studies focused on tumor and cancer detection. Advanced image processing and object detection methods can enhance detection accuracy. Image segmentation, crucial for image analysis, pattern recognition, and medical imaging, divides an image into regions of homogeneous characteristics, such as intensity, color, or texture. Techniques include Threshold, Edge, Region-Based, Clustering-Based, and Artificial Neural Network-Based segmentation. Edge-based segmentation can be performed with operators like Sobel, Canny, Prewitt, Laplacian of Gaussian, and Roberts operators [5]. Over recent years, numerous approaches have been proposed for extracting features from bone tumor images, essential for classifying tumors. Feature extraction methods include statistical, structural, model-based, transformer, and deep learning techniques [6] [7]. Classification, a key step, may be binary or multiclass, utilizing techniques like Support Vector Machines (SVM), Decision Trees, and Convolutional Neural Networks (CNNs) [8]. Traditional segmentation and feature extraction methods (e.g., binary thresholding, blob clustering, edge detection, and fuzzy C-means) have been used but lack automation, often resulting in inconsistent accuracy. Deep learning, especially CNNs, offers automated feature engineering, interpretability, and reproducibility, enhancing tumor detection and classification [9]. Object detection algorithms are typically classified into one-stage (faster but less accurate) and two-stage methods (slower but more precise) [10]. While many object detection models are optimized for general images, medical images require specific adaptation [11][12]. Previous studies have demonstrated classification capabilities but often lack robust tumor localization [13]. Our study addresses this gap by combining classification and precise localization using Faster R-CNN, offering a more comprehensive solution. Our model incorporates advanced techniques like data augmentation, transfer learning, and early stopping to reduce overfitting and improve generalization. Faster R-CNN leverages Region Proposal Networks (RPNs) alongside Fast R-CNN, achieving effective localization and classification but at a computational cost [14]. In our approach, classification (recognition) and localization (detection) are performed simultaneously. We build a CNN model from scratch as the framework's backbone, applying transfer learning for classification and RPNs for bounding box generation.

II. Methodology

The proposed framework for "Bone Tumor Detection using Faster R-CNN" as seen in Fig. 1 begins by loading X-ray images from a dataset containing both normal and tumor-affected bone samples. During preprocessing, images are standardized, converted to grayscale, and denoised using Median and Gaussian filters. Categorical features are one-hot encoded, with 0 for 'tumor' and 1 for 'non-tumor'. The dataset

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is split into training, testing, and validation sets, and a CNN model is developed from scratch. Data augmentation and early stopping are applied during training to prevent underfitting and overfitting.

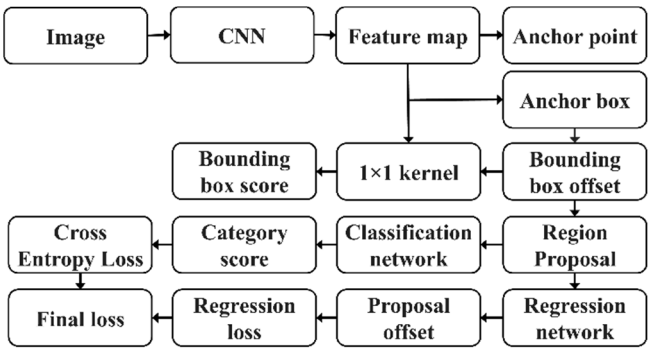

Fig. 1. Proposed Faster R-CNN model

Input images are processed through a CNN backbone to create feature maps where anchor boxes are projected. A 1×1 convolution predicts categories and offsets for all anchor boxes. During training, anchor boxes are classified as positive (with significant overlap with ground truth boxes) or negative (minimal overlap). Positive anchors are assigned objects, and negatives are assigned background. Classification uses cross-entropy loss, while offsets are learned with L2 regression loss to align positive anchors more closely with ground truth boxes. These adjusted anchors serve as region proposals, processed in stage-1 by the Region Proposal Network (RPN).

In stage-2, the RPN's region proposals are resized via Region of Interest (ROI) pooling and passed through another convolutional network. This network refines object categories (trained with cross-entropy loss) and further adjusts offsets (using L2 regression loss) to match ground truth boxes. The combined losses in a weighted approach enable multi-task learning. During inference, high-scoring boxes from stage-1 are refined in stage-2 for category and offset accuracy. Finally, non-max suppression eliminates duplicate boxes, ensuring precise object detection.

A. Preparing Data & Data Loader in Pytorch

Images are labeled using 'CVAT' software. A PyTorch Data Loader handles shuffling, sampling, and batching.

B. Passing Through the Backbone Network

This process involves obtaining a pre-trained model, creating a base model, freezing layers, adding new trainable layers, training these layers, and fine-tuning the model. This process results in feature-rich representations (feature maps) of the images.

C. Anchor Points and Boxes Generating

Each anchor point in the feature map represents specific coordinates in the width and height dimensions. For each anchor point, nine bounding boxes of various shapes and sizes are generated to encompass all objects in the image, tailored according to the dataset. The Intersection over Union (IoU) is calculated between the generated anchor boxes and projected ground truth boxes, aligning each ground truth box to its closest grid point in the feature space. Anchor boxes with IoU ≥ 0.7 are labeled as positive, while those with IoU ≤ 0.3 are labeled as negative. This process is illustrated in Fig. 2.

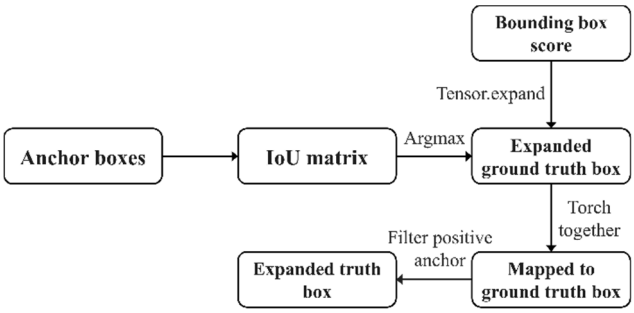

Fig. 2. Filtering positive samples

To improve detection accuracy, a Python function calculates offsets between positive anchor boxes and ground truth boxes. The function flattens the batch, filters positive samples, and produces both positive and negative anchor boxes, as shown in Fig. 3.

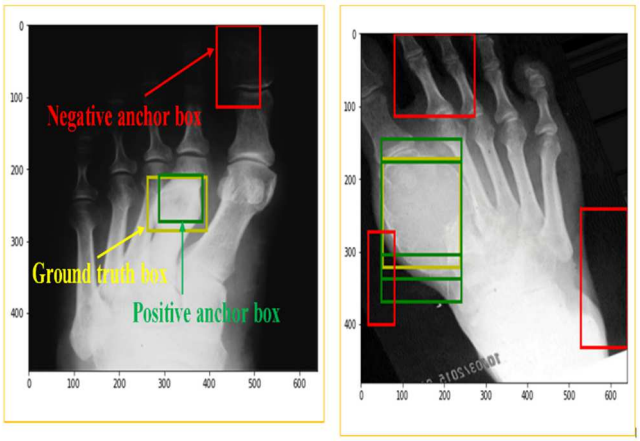

Fig. 3. Positive and Negative anchor boxes.

D. Region Proposal Network

The Region Proposal Network (RPN) consists of three main components: a backbone network, a sampling module, and a proposal module. Its primary function is to classify each anchor box as either an object or background and predict offsets for localization using two 1x1 convolutional layers. One layer calculates classification scores to label each anchor box, while the other predicts the scores and offsets from the feature map. Positive anchor boxes are selected, and the predicted offsets are applied to generate region proposals during training.

The equations used for region proposal calculations are outlined in Equations (1-4).

$$x_c^p = x_c^a + t^x w^a \tag{1}$$

$$y_c^p = y_c^a + t^y h^a \tag{2}$$

$$w^p = w_a \exp(t^w) \tag{3}$$

$$h^p = h_a \exp(t^h) \tag{4}$$

Here,
x = coordinate of anchor box along the X-axis
y = coordinate of anchor box along the Y-axis
p = region proposal

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c = center of the anchor box
h = height of the anchor box
a = number of anchor boxes
t = predicted offsets.

The RPN produces scores and offsets for all anchor boxes during training and inference. The classification loss is computed using positive and negative anchor boxes, while the L2 regression loss is calculated using positive sample offsets. Proposals are generated using predicted offsets during inference, and the raw model logits are converted to probability scores using the sigmoid function

E. Classification Module

A customized convolutional network predicts object categories within region proposals, which are first normalized in size using ROI pooling. Each proposal is processed through a CNN with a convolutional layer, an average pooling layer, and a linear layer to produce category scores. The SoftMax function identifies the category with the highest probability score during inference, with cross-entropy used for classification loss during training. Additionally, a regression network refines proposal offsets, and duplicate bounding boxes are removed using non-max suppression in the final step.

F. Training the model

The dataset used to train the model contains 3,522 images, with 2,283 showing tumors and the rest classified as normal. Training uses a standard loop with the Adam optimizer, set at a learning rate of 0.0001.

III. RESULT ANALYSIS & DISCUSSION

Our approach to detecting bone tumors involves two main steps: recognition and localization. First, a CNN model, serving as the backbone, is trained on our dataset to recognize the presence of tumors and output corresponding probabilities. Next, the Region Proposal Network (RPN) identifies the tumor's location by generating bounding boxes. The model's predictions show the likelihood of a tumor's presence or absence, as illustrated in Fig. 4. It displays the predicted probability of tumor presence or absence with confidence.

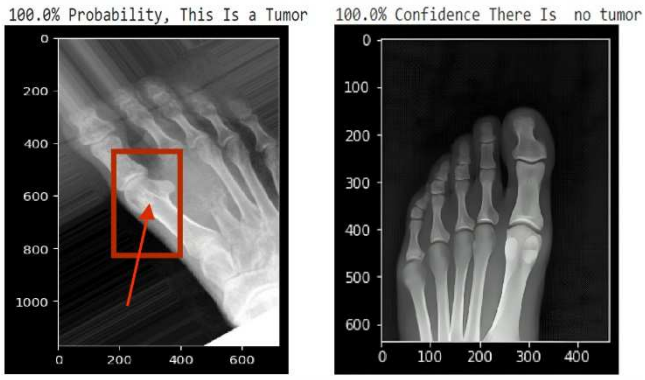

Fig. 4. Classification Output of the CNN Model

The test and validation losses of the CNN model for different numbers of epochs have been shown in Fig.5. The both losses decrease with the increases in epoch numbers. When the CNN model recognizes tumors, the RPN module locates the position of the tumors in the second step. The detection of the tumor location is shown in Fig.6.

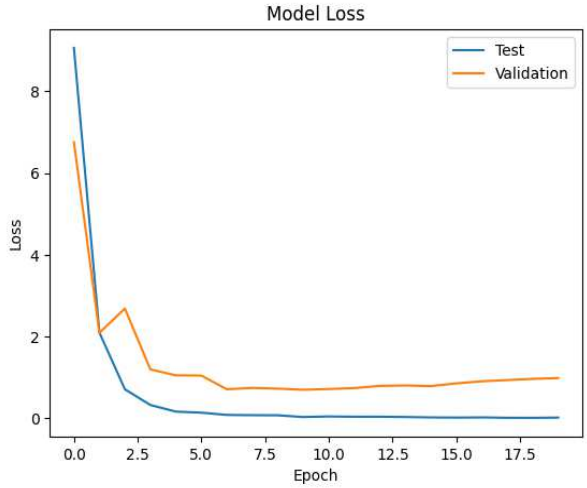

Fig. 5. Loss vs. epoch graph

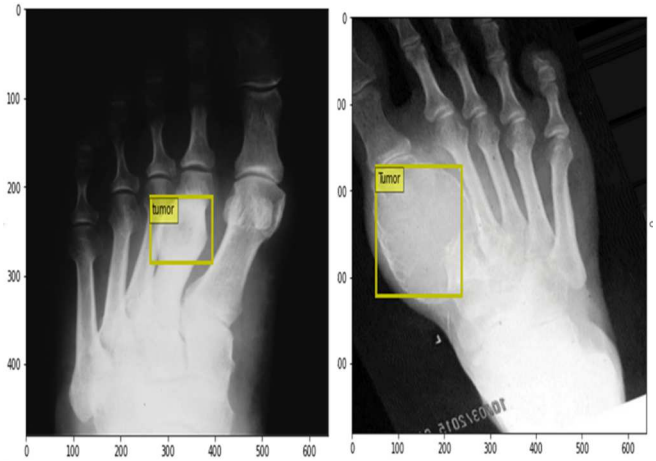

Fig. 6. Detection of tumor

The performance of the model in terms of localizing the tumor's position is evaluated by the mean average precision (mAP) and the mean IoU of the model is shown in Fig.7. As the number of proposals increases, both mAP and mean IoU increase to a certain level rapidly and then the progress gets slower. Hence, for a specific number of proposals, the best performance is achieved. In the Faster R-CNN model with 300 proposals from the region proposal network, we observed that reducing the number of proposals improves inference time at the cost of slight reductions in precision and recall. We assessed precision, recall, and mean IoU across varying numbers of proposal boxes, revealing minimal performance variation between 280 and 300 proposals where 96.28% mAP is achieved. Mean average precision and IoU are closely aligned in this configuration.

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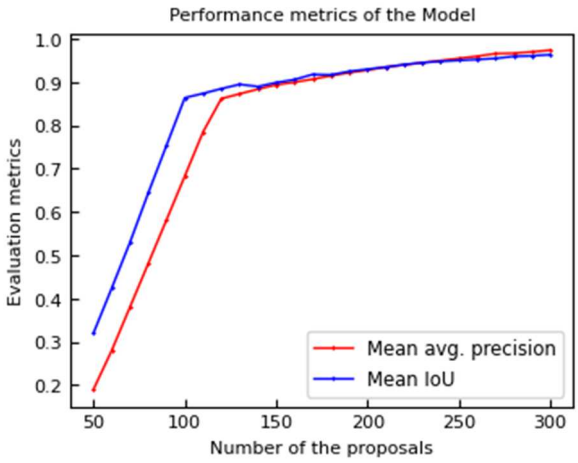

Fig. 7. Performance metrics of the RPN

IV. COMPARISON WITH STATE-OF-THE-ART METHODS

A performance comparison of the proposed ensemble method against existing methods is presented in Table I, highlighting studies with various models and encouraging outcomes. However, differing architectures, training datasets, class numbers, and design parameters make direct comparisons unjust.

Ref.	Model	Accuracy	Classification	Detection
[15]	ANN	92.50%	✓	×
[16]	Fuzzy C-mean clustering.	93.75%	✓	×
[17]	CNN	94.5%	✓	×
This work	Faster R-CNN	98.73%	✓	✓

V. FUTURE SCOPE

- The size of the detected tumor can be calculated by improving the algorithm.
- It can be commercialized for real-time detection with more inspection and by developing an Android application or software.
- The model can be tested with different datasets to evaluate and compare the performance.

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